

ON THE CONTROL EFFICIENCY OF MULTI-AGENT LANGUAGE MODEL SYSTEMS

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ABSTRACT

How efficiently can an external controller steer a group of interacting language-model agents? We study this question in relational reasoning tasks where evidence is distributed across the population. Our feedback controller has two separate limits: it observes only a sample of the agents and can intervene in only a fixed number of their interactions. This setup lets us measure the distinct effects of sensing and actuation. We introduce finite-time measures for three aspects of control: how much an intervention changes the population, how efficiently action-dependent information becomes coherent population motion, and how efficiently the finite-time thermodynamic control budget becomes directed current. We derive a finite-state reference theory with a single measurable directional affinity h and a kinetic compliance γ , which separates which direction controlled revisions favor from how readily agents revise at all. In experiments with matched initial social and knowledge states, advocacy increases support for the controller’s target in every tested condition, with susceptibilities from 0.100 to 0.163. The transmitted control information depends strongly on the population state. More actuation produces a wider informative region, whereas stronger feedback gain is predicted to concentrate information near the controller’s decision boundary. Information–response efficiency increases with actuation across the full resource grid, from about 0.4% to 2.5% of the Pinsker upper bound. A calibrated finite-compliance model also reproduces the observed controlled current closely ($r = 0.999$). Finally, the single-affinity theory yields the exact finite-time identity $\Delta S_{\text{sys}} + hJ_c + I(n_k; Y_k) = \Sigma \geq 0$ and the bounded thermodynamic efficiency $\eta_{\text{th}} = hJ_c/[hJ_c + I(n_k; Y_k)]$. The resulting theory predicts diminishing returns from additional sensing whenever the acquired information does not produce a proportional increase in directed current. Our framework measures control efficiency while keeping it separate from whether the agents reach the correct answer.

1 INTRODUCTION

Large language models (LLMs) are increasingly deployed not only as individual problem solvers, but as components of *multi-agent systems* in which many autonomous agents communicate, exchange information, and collectively determine an outcome. Recent work has shown that populations of LLM agents can exhibit genuinely collective phenomena, including spontaneous convention formation, amplification of population-level biases, and higher-order coordination that cannot be understood from isolated agents alone (Ashery et al., 2025; Riedl, 2026). These observations raise a complementary question that becomes increasingly important as such systems are deployed: *how can an external agent control a collective of interacting LLMs?* In particular, we are interested not in unconstrained steering—where a controller can inspect and modify the entire system—but in the more realistic regime in which the controller itself has limited resources.

We study this problem through a *bounded feedback controller*. At each population round, the controller observes only a finite random sample of agents and must use this partial observation to decide whether to intervene. Its sensing capacity is therefore limited: the controller never observes the complete population state and its estimate is intrinsically noisy. Its actuation capacity is limited as well: even when it decides to act, it may replace the social input in only a bounded number of interactions. The resulting closed loop has the form

$$N_t \longrightarrow Y_t \longrightarrow U_t \longrightarrow N_{t+1}, \quad (1)$$

where N_t denotes the population state, Y_t the controller’s partial observation, and U_t its intervention. This formulation makes imperfect information a consequence of finite sensing resources rather than an artificial source of observation noise, and separates two distinct constraints on control: *how much of the system can be observed* and *how much of it can be acted upon*. Our central question is therefore:

How efficiently can a resource-constrained feedback controller steer an interacting population of LLM agents when both information and actuation are limited?

Controlling an LLM population, however, should involve more than controlling a population of stochastic finite-state machines. The distinctive feature of language-model agents is precisely that their interactions can carry *semantic information*: agents can possess different facts, communicate those facts through language, accumulate evidence, and reason over combinations of information. We therefore introduce an explicit *epistemic load* into the collective dynamics using relational reasoning tasks with distributed evidence. The information required to solve each task is spread across the population so that the collective contains a complete proof while individual agents initially do not. Each agent consequently carries two coupled states: a social state, represented by its current answer or vote $X_i(t)$, and an epistemic state, represented by the set of supporting facts $K_i(t)$ available to it. Local language interactions can change either or both. An agent may adopt another agent’s answer through social influence without acquiring the evidence that supports it, while an explicitly communicated fact may expand its knowledge and subsequently alter its reasoning. This distinction allows us to study how *social influence, information propagation, and reasoning jointly determine controllability*.

This separation is particularly important for feedback control. In our main setting, the controller communicates only a recommendation and does not provide task evidence. It can therefore directly perturb the social dynamics but cannot directly increase the epistemic content of the population. We can consequently distinguish whether an intervention changes what agents *believe* from whether it changes what they *know*. Moreover, by allowing the controller target to disagree with the objectively correct answer, controllability and task performance become distinct quantities: a controller can successfully steer the population while simultaneously moving it away from truth. This provides a controlled setting for asking when accumulated evidence makes a population resistant to external influence, when social pressure overwhelms reasoning, and whether the epistemic state of a collective changes the amount of control required to redirect it.

To quantify these processes, we treat sensing and actuation as two distinct information channels. Writing n_k for the current number of agents that support the controller target, Y_k for the controller’s noisy observation of that count, U_k for the action selected from that observation, and n_{k+1} for the target-support count after actuation, the two quantities are

$$I(n_k; Y_k) \quad (\text{sensing information}), \quad (2)$$

$$T_\pi(n) \equiv I(U_k; n_{k+1} \mid n_k = n) \quad (\text{action-to-population information}). \quad (3)$$

The first measures information acquired by the sensor about the current population state. The second measures, at a fixed current state, how much the controller action is expressed in the population state after the actuation step; the subscript π indicates that this channel is induced by the chosen feedback policy. We define π and the complete control loop below. These quantities are complemented by signed response statistics, population currents, and their fluctuations. We further introduce a finite-state feedback reference with one measurable directional affinity h and a kinetic compliance γ . The affinity determines the preferred direction of a controlled revision, whereas γ determines how readily a controlled opportunity produces any revision. This separation is directly calibratable from microscopic transitions and avoids introducing distinct “controller” and “epistemic-load” affinities that cannot be independently identified from the controlled channel.

This perspective connects naturally to feedback thermodynamics, where measurement and control are treated as coupled information-processing operations (Horowitz & Sandberg, 2014), and to recent stochastic-thermodynamic formulations of social imitation, where probability currents, dynamical activity, entropy production, and uncertainty relations characterize collective dynamics without requiring a literal energetic interpretation (Irisarri et al., 2026). In our finite-time feedback layer, the same single affinity leads to an exact path-space identity, $\Delta S_{\text{sys}} + hJ_c + I(n_k; Y_k) = \Sigma$, with Σ the KL irreversibility between forward and reverse feedback paths. Importantly, thermodynamics is not the motivation for introducing the controller, and Σ is not interpreted as literal energetic

dissipation of the language models. Rather, it provides a principled language for characterizing the *resource–response trade-offs* generated by the bounded feedback problem: how much information is acquired, how much directed current is produced, how irreversible the resulting coarse-grained feedback cycle is, and where additional control resources cease to be useful.

Finally, we introduce a matched finite-population q -voter process as a classical reference model. The reference system is subjected to the same population size, local interaction structure, finite sensing process, stochastic feedback policy, and bounded actuation schedule as the LLM population. It therefore captures the response expected from a system whose collective dynamics arise from stochastic local imitation alone. This comparison is deliberately not intended to claim that LLM agents are q -voters. Instead, the q -voter provides a controlled null model in which semantic communication, distributed evidence, and language-based reasoning are absent. Agreement with this reference identifies behavior that can already be explained by ordinary stochastic social influence; systematic departures reveal where the epistemic and reasoning capabilities of language-model agents modify the control dynamics.

Taken together, our goal is to develop a quantitative account of *control efficiency in reasoning multi-agent systems*: how limited sensing and limited actuation are converted into directed changes of a collective state, and how this conversion is altered when the interacting agents communicate meaningful information and reason with it. Our main contributions are:

- We formulate control of an LLM collective as a *resource-constrained feedback problem*, explicitly separating finite sensing capacity from finite actuation capacity and treating imperfect state information as a consequence of bounded observation.
- We introduce a relational multi-agent environment with distributed evidence that exposes both *social* and *epistemic* population dynamics, allowing us to distinguish changes in expressed belief from changes in the information available for reasoning.
- We develop information-theoretic and stochastic-dynamical observables that quantify sensing, transmission of controller actions, directional response, currents, fluctuations, and finite-time path irreversibility. From the same accounting we derive a bounded single-affinity thermodynamic efficiency that measures how much of the finite-time sensing–actuation expenditure is converted into directed current.
- We construct a matched finite- N q -voter reference under the same feedback protocol, providing a classical imitation baseline against which the specifically language- and reasoning-dependent behavior of the LLM population can be identified.

2 RELATED WORK

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3 BACKGROUND

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4 EXPERIMENTAL SYSTEM: TASKS AND THE BLACKBOARD GAME

Our experiment separates task construction from collective decision-making. We first create a team-allocation problem with a known answer and divide its evidence among language-model agents. We then let the agents exchange that evidence and revise their answers through a shared message board. A controller can post on the same board to encourage a chosen answer. The resulting game lets us measure how communication changes both the information available to agents and their collective decision.

4.1 CONSTRUCTING THE TASK

A team-allocation problem. Our tasks adapt the Team Allocation setting of MuSR (Sprague et al., 2024), which tests multistep reasoning from natural-language evidence. Three people must be assigned to two jobs: one person works alone on the first job, and the remaining pair works together on the second. There are therefore three possible answers. For example, suppose the jobs are to analyze field measurements and organize a community workshop. The choices are:

Analyze measurements	Organize the workshop
Alice	Bruno and Chandra
Bruno	Alice and Chandra
Chandra	Alice and Bruno

Choosing among them requires evidence about individual skills and cooperation within each possible pair. We generate our own instances of this task family.

Establishing the answer before writing the evidence. We first construct a hidden numerical model with six skill values (one per person and job) and three pairwise cooperation values. Each value lies

in $\{1, 2, 3\}$. If person p takes the first job and people u, v take the second, our allocation score is

$$S(p; u, v) = s_{p,1} + s_{u,2} + s_{v,2} + c_{u,v}, \quad (4)$$

where $s_{p,t}$ measures person p 's skill for job t and $c_{u,v}$ measures cooperation. We evaluate all three allocations and retain worlds with a unique highest-scoring answer. Thus, the correct answer is determined by code before any language-model interaction. Agents receive the scenario and evidence, while the numerical model and scores remain hidden.

Turning the hidden model into evidence cards. Following MuSR's approach of constructing a reasoning problem before rendering it in language (Sprague et al., 2024), we create short accounts of behavior that support facts about skills or cooperation. For example, an account of Alice identifying and repairing a spreadsheet error provides evidence of her analytical ability. We call each short account an *evidence card*. Every card is linked to a proposition that can be checked against the hidden model, such as a person's skill level or a comparison between two people's skills. An LLM writes the accounts; equality comparisons use fixed templates. We check that the text supports its intended proposition and does not disclose the hidden scores or recommend an answer. This preserves the distinction between the exact answer, established by the numerical model, and the natural-language reasoning needed to infer it from the cards.

Distributing incomplete information. We assign one private card to each of $N = 24$ agents. A single card leaves several answers plausible, whereas combining cards provides stronger evidence. For example, one agent may know about Alice's analytical ability and another about Bruno and Chandra's cooperation. We assess this partial-information ambiguity by enumerating the numerical models consistent with each private card and checking which allocation wins in each model. The assignment includes complementary cards that favor the correct answer when considered together. Agents may initially guess correctly, but they must exchange evidence to access information beyond their own card.

We freeze the task and evidence before play. Each agent first votes using only its private card. These initial votes are saved and reused across matched experimental conditions. For each model call, we shuffle the A/B/C labels assigned to the three allocations and translate the response back to the chosen allocation, preventing a fixed answer letter from becoming a shared social cue.

4.2 PLAYING THE BLACKBOARD GAME

The *blackboard* is a shared collection of short-lived messages. Agents take turns reading a small random sample of posts, choosing an answer, and optionally adding a post of their own. A message posted early can therefore influence later decisions within the same round. For example, an agent may post the card about Alice's analytical ability; another agent who reads it can combine it with its own cooperation evidence and reconsider the allocation. Access to the board is limited: agents see only the messages they sample.

Agent state and memory. Agent i maintains a current answer X_i , an active evidence set K_i , and a historical set H_i containing every card it has received. Only active evidence is available in its prompt. At each round boundary, each active card survives independently with probability ρ ; historical records are retained. Reading a previously lost card makes it active again. The parameter ρ therefore controls how long communicated evidence remains usable. Previous private explanations are recorded for analysis but are not fed back to agents.

One round. Each round has a boundary stage, a controller posting stage, and N sequential agent updates. The controller first samples votes and decides whether to act. The previous round's messages then expire, evidence persistence is applied, and an active controller posts its messages. During each subsequent update:

1. One focal agent is sampled uniformly from the population. It reads up to q live board messages, sampled uniformly without replacement, excluding its own posts.
2. The agent receives the question, answer options, active evidence, previous vote, and sampled messages. One LLM call returns an updated vote, a private explanation, and an optional public message.

3. The vote is updated immediately, attached evidence from the sampled reports is added to the agent’s active and historical sets, and any new public message becomes available to later updates in the same round.

Focal agents are sampled independently at each position, so a round need not update every agent once. Messages live only for their creation round; an empty board supplies no social messages. Episodes run for a fixed number of rounds, including after consensus.

Messages and evidence transmission. An ordinary agent may post a `REPORT`, ask for evidence with a `REQUEST`, or post nothing. A report contains public text and may attach one evidence identifier available to its author. Requests carry no evidence. Posts also carry the author’s current allocation choice, and replies may refer only to messages visible in that update. The prompt distinguishes a participant’s interpretation from the canonical evidence text attached to a report. Only reading a report with a valid attachment changes the exact evidence state. Posting alone does not transmit a card to anyone, and public prose is not counted as verified evidence. This separates the transmission of task evidence from persuasion through an agent’s interpretation of that evidence.

Feedback through the same public board. Let Z be the controller’s target, which may be correct or incorrect, and let

$$n_k = \sum_{i=1}^N \mathbb{I}[X_i(k) = Z], \quad x_k = n_k/N \quad (5)$$

be its support before round k . The controller samples q_c distinct agents and observes their votes, giving a target-support count Y_k . The binary feedback policy in Section 5 chooses between silence ($U_k = 0$) and communication ($U_k = 1$); lower sampled target support makes communication more likely. This decision uses votes only.

The controller has a fixed pool of true evidence cards. To favor an incorrect answer, it selectively discloses evidence that keeps that answer plausible while withholding evidence that would weaken it. Its reports remain truthful at the fact level.

On active rounds, an adaptive communication policy chooses to publish up to b evidence reports, make one request for evidence, or post one `DIRECTIVE` directing attention to a comparison. The latter two modes carry no evidence. An LLM selects the mode and eligible report identifiers using sampled votes, previous public messages, the allowed fact pool, and posting history. It has no access to agents’ private evidence or explanations. The selected reports reproduce the frozen evidence text, while requests and directives use fixed templates. This communication choice occurs only after the binary decision to act. A report-only variant instead posts exactly b distinct evidence reports on each active round.

The controller appears under a stable participant identity. Its messages enter the same board and have no sampling priority. Consequently, b bounds posting capacity, while actual exposure depends on which messages agents read. We record posts, reads, evidence acquisition and reactivation, and vote changes separately. Tracking support for both Z and the correct answer, alongside active and historical evidence, allows us to distinguish successful steering from improved task performance.

5 EFFICIENCY OF FEEDBACK CONTROL

We use a solvable blackboard reference to separate sensing, message exposure, and behavioral response. Let $n_k \in \{0, \dots, N\}$ count supporters of the controller target Z before round k , and write $x_k = n_k/N$. One feedback cycle has causal order $n_k \rightarrow Y_k \rightarrow U_k \rightarrow n_{k+1}$, where $U_k = 0$ denotes silence and $U_k = 1$ active posting. Both branches contain M population updates, usually $M = N$. Posting b messages changes the chance of exposure; it does not guarantee b controlled revisions. The results below are exact for the reference kernels, rather than a fitted description or an energetic thermodynamics of the full language-model system.

5.1 SENSING, POSTING, AND SAMPLED EXPOSURE

The controller senses q_c agents without replacement. For $Y = y$ sampled target supporters, the sensor and soft feedback policy are

$$S(y | n) = \frac{\binom{n}{y} \binom{N-n}{q_c-y}}{\binom{N}{q_c}}, \quad (6)$$

$$\pi(1 | y) = \sigma \left[\beta \left(\theta - \frac{y}{q_c} \right) \right], \quad \sigma(z) = (1 + e^{-z})^{-1}, \quad (7)$$

$$a_n = P(U = 1 | n) = \sum_y S(y | n) \pi(1 | y). \quad (8)$$

Suppose the frozen eligible board contains B peer messages and, when active, b controller messages. Each focal update samples $s = \min(q, B + b)$ messages uniformly without replacement. Its probability of seeing at least one controller message is

$$w_b = 1 - \frac{\binom{B}{s}}{\binom{B+b}{s}}. \quad (9)$$

Here $\binom{B}{s} = 0$ when $s > B$. The theory uses the number of messages *actually posted*; a nominal budget must be replaced by the realized count when fewer messages are posted. Thus q_c, θ, β govern the posting decision, while B, b, q govern exposure. For positive q , exposure saturates as b grows.

5.2 TWO RESPONSE CLASSES AND EXACT ROUND KERNELS

An update with compliance $\gamma \in [0, 1]$ and conditional target preference $p \in [0, 1]$ follows

$$K_{\gamma,p}(n+1 | n) = \gamma(1 - n/N)p, \quad (10)$$

$$K_{\gamma,p}(n-1 | n) = \gamma(n/N)(1-p), \quad (11)$$

$$K_{\gamma,p}(n | n) = 1 - K_{\gamma,p}(n+1 | n) - K_{\gamma,p}(n-1 | n). \quad (12)$$

A revision opportunity can leave the vote unchanged. Let $K_0 = K_{\gamma_0, p_0}$ describe unexposed updates and $K_1 = K_{\gamma_1, p_1}$ exposed updates. Marginalizing exposure gives the exact closure

$$K_b = (1 - w_b)K_0 + w_b K_1 = K_{\gamma_b, p_b}, \quad (13)$$

$$\gamma_b = (1 - w_b)\gamma_0 + w_b\gamma_1, \quad p_b = \frac{(1 - w_b)\gamma_0 p_0 + w_b\gamma_1 p_1}{\gamma_b}. \quad (14)$$

When $\gamma_b = 0$, K_b is the identity and p_b is immaterial. The silent and active *round* kernels are

$$Q_0 = K_0^M, \quad Q_1 = K_b^M, \quad Q_\pi(m | n) = (1 - a_n)Q_0(m | n) + a_n Q_1(m | n). \quad (15)$$

These powers assume a frozen board, homogeneous response parameters, binary exposure, and a shared unexposed baseline across branches. In particular, Q_0 generally evolves the population even when the controller is silent.

5.3 RAW AND AVAILABILITY-NORMALIZED SUSCEPTIBILITY

At a common initial state, the primary response observable is

$$\chi_b(x) = \mathbb{E}[x' | U = 1, x] - \mathbb{E}[x' | U = 0, x]. \quad (16)$$

The one-update mean obeys $\mathbb{E}[x_{j+1} | x_j] = (1 - \gamma/N)x_j + \gamma p/N$. Consequently the exact round mean and susceptibility are

$$F_M(x; \gamma, p) = p + (x - p)(1 - \gamma/N)^M, \quad (17)$$

$$\begin{aligned} \chi_b(x) &= F_M(x; \gamma_b, p_b) - F_M(x; \gamma_0, p_0) \\ &= p_b(1 - r_b) - p_0(1 - r_0) + (r_b - r_0)x, \end{aligned} \quad (18)$$

where $r_b = (1 - \gamma_b/N)^M$ and $r_0 = (1 - \gamma_0/N)^M$. The reference predicts an affine dependence on initial target share, with budget entering through exposure. If $r_b \neq r_0$, the zero-response point is $x_b^* = [p_0(1 - r_0) - p_b(1 - r_b)]/(r_b - r_0)$; it need not lie in $[0, 1]$. When $\gamma_0 = \gamma_1 = \gamma$, the state dependence cancels:

$$\chi_b(x) = w_b(p_1 - p_0) [1 - (1 - \gamma/N)^M]. \quad (19)$$

Exposure and compliance therefore impose distinct limits: being seen more often and changing behavior after exposure are separate mechanisms.

To express the same branch contrast relative to the remaining non-target population, define

$$\chi_{b,\text{avail}}(x) = \frac{\chi_b(x)}{1 - x}, \quad x < 1. \quad (20)$$

The saturated state $x = 1$ is excluded. This diagnostic can be negative or exceed one: the contrast includes differential retention of existing target supporters as well as additional adoption. It is neither a conversion probability nor a bounded efficiency. For occupancy $p(n)$, the corresponding available-mass-weighted summary is

$$\bar{\chi}_{\text{avail}}[p] = \frac{\sum_{n=0}^{N-1} p(n) \chi_b(n/N)}{\sum_{n=0}^{N-1} p(n) (1 - n/N)}, \quad (21)$$

when the denominator is positive. Both sums exclude saturation; this ratio avoids an unweighted average of potentially large near-saturation ratios.

5.4 ACTION INFORMATION AND INFORMATION-RESPONSE EFFICIENCY

The exact state-local action-to-population information is the weighted Jensen–Shannon divergence of the two evolving round kernels:

$$T_\pi(n) = I(U; n' | n) = \sum_u P(u | n) \sum_m Q_u(m | n) \log_2 \frac{Q_u(m | n)}{Q_\pi(m | n)}. \quad (22)$$

With $H_2(a) = -a \log_2 a - (1 - a) \log_2 (1 - a)$, $0 \leq T_\pi(n) \leq H_2(a_n)$. Thus the action-information fraction is

$$\eta_{\text{IF}}(n) = \frac{T_\pi(n)}{H_2(a_n)} \in [0, 1], \quad (23)$$

when $H_2(a_n) > 0$. It measures how much conditional action uncertainty is visible in the next state, rather than directional motion.

Because $x' \in [0, 1]$, Pinsker’s inequality gives

$$T_\pi(n) \geq B_{\text{IR}}(n) \equiv \frac{2a_n(1 - a_n)}{\ln 2} \chi_b(n/N)^2. \quad (24)$$

The bounded information–response efficiency is therefore

$$\eta_{\text{IR}}(n) = \frac{2a_n(1 - a_n) \chi_b(n/N)^2}{(\ln 2) T_\pi(n)} \leq 1, \quad (25)$$

for $T_\pi(n) > 0$. This bound uses the *raw* susceptibility, not its availability-normalized counterpart, and does not require Q_0 to be the identity. An informative action channel can have little mean response if its effect is mainly on fluctuations or higher moments.

5.5 EXCESS CURRENT AND FINITE-TIME PATH ACCOUNTING

Let $p_k(n)$ denote the pre-round ensemble and $\mu_u(n) = \mathbb{E}[n' - n | n, U = u]$. Controller-excess current subtracts the silent counterfactual:

$$J_{c,k}^{\text{ex}} = \sum_n p_k(n) a_n [\mu_1(n) - \mu_0(n)] = N \sum_n p_k(n) a_n \chi_b(n/N). \quad (26)$$

Equivalently, $J_{c,k}^{\text{ex}} = J_{\text{tot},k} - J_{\text{sil},k}$, where $J_{\text{tot},k} = \sum_n p_k(n)[(1 - a_n)\mu_0(n) + a_n\mu_1(n)]$ and $J_{\text{sil},k} = \sum_n p_k(n)\mu_0(n)$.

For interior preferences, define $h_0 = \ln[p_0/(1 - p_0)]$ and $h_b = \ln[p_b/(1 - p_b)]$, with $h_u = h_0$ under silence and $h_u = h_b$ under active posting. On supported transitions, weighted reversibility gives

$$\ln \frac{Q_u(m | n)}{Q_u(n | m)} = \ln \binom{N}{m} - \ln \binom{N}{n} + h_u(m - n). \quad (27)$$

Let $p_{Y,k}(y) = \sum_n p_k(n)S(y | n)$ and let p_{k+1} be the ensemble propagated by Q_π . The forward and reverse-reference path measures are

$$P_F(n, y, u, m) = p_k(n)S(y | n)\pi(u | y)Q_u(m | n), \quad (28)$$

$$P_R(m, y, u, n) = p_{k+1}(m)p_{Y,k}(y)\pi(u | y)Q_u(n | m). \quad (29)$$

Their log ratio yields the exact finite-time identity

$$\Sigma_k = D_{\text{KL}}(P_F \| P_R) = \Delta S_{\text{sys},k} + I(n_k; Y_k) + \mathcal{A}_k \geq 0, \quad (30)$$

where all terms use natural logarithms and

$$S_{\text{sys}}[p] = - \sum_n p(n) \ln p(n) + \sum_n p(n) \ln \binom{N}{n}, \quad (31)$$

$$\mathcal{A}_k = \sum_n p_k(n) [(1 - a_n)h_0\mu_0(n) + a_nh_b\mu_1(n)]. \quad (32)$$

The entropy change is retained; stationarity is not assumed.

For $\mathcal{A}_k \geq 0$ and positive denominator, the bounded *total path-transport fraction* is

$$\eta_{\text{path},k} = \frac{\mathcal{A}_k}{\Sigma_k - \Delta S_{\text{sys},k}} = \frac{\mathcal{A}_k}{\mathcal{A}_k + I(n_k; Y_k)} \in [0, 1]. \quad (33)$$

It includes autonomous baseline transport and is not a controller-only thermodynamic efficiency. Indeed, with $J_{\text{adv},k} = \sum_n p_k(n)a_n\mu_1(n)$,

$$\mathcal{A}_k - h_0J_{\text{sil},k} = h_0J_{c,k}^{\text{ex}} + (h_b - h_0)J_{\text{adv},k}. \quad (34)$$

This excess-affinity diagnostic need not be nonnegative and is not itself a second-law-bounded expenditure. A sufficient attribution condition is a neutral baseline affinity, $p_0 = 1/2$ or $h_0 = 0$. Only under this additional condition does the path fraction become

$$\eta_{\text{th},k}^{(h_0=0)} = \frac{h_bJ_{\text{adv},k}}{h_bJ_{\text{adv},k} + I(n_k; Y_k)} \in [0, 1], \quad (35)$$

provided $h_bJ_{\text{adv},k} \geq 0$ and the denominator is positive. Neutral affinity does not imply zero silent drift. In general, the pair $(J_{c,k}^{\text{ex}}, \Sigma_k)$ describes excess steering versus total coarse path irreversibility without assigning baseline transport to the controller.

Empirical scope. Exposure frequency and exposed/unexposed transition rates can test the reference. If $p_{+,r}$ and $p_{-,r}$ are conditional target-entry and exit probabilities in response class r , then $\gamma_r = p_{+,r} + p_{-,r}$ and $p_r = p_{+,r}/(p_{+,r} + p_{-,r})$ when the denominator is nonzero. Shared parameters should predict held-out budgets and states; observational exposure classes alone do not establish a causal effect. Paired active and silent continuations should match the complete state, including board contents and epistemic information. Board growth, self-exclusion, message content, and heterogeneous epistemic state can violate the frozen, homogeneous reference. Such departures call for explicit conditioning or time-dependent kernels. The safe bounded information quantities are η_{IF} and η_{IR} ; controller-specific thermodynamic attribution requires additional assumptions.

6 RESULTS

We organize the results around three complementary questions. First, how much can the controller move the population? Second, how efficiently is the controller-dependent information expressed as coherent population motion? Third, how efficiently is the finite-time thermodynamic control budget converted into directed current? These questions correspond respectively to the susceptibility χ , the information–response efficiency η_{IR} , and the thermodynamic efficiency η_{th} .

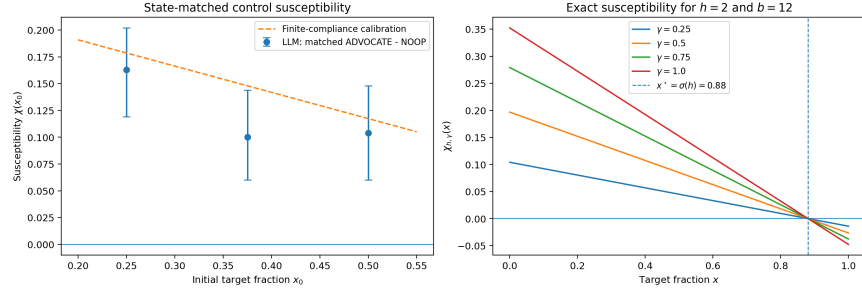


Figure 1: **Empirical and theoretical susceptibility.** (a) State-matched ADVOCATE-minus-NoOp response with 95% bootstrap intervals and the finite-compliance calibration from microscopic transition rates. (b) Exact theory showing how kinetic compliance γ changes the response amplitude without moving the intrinsic set point.

6.1 CONTROL SUSCEPTIBILITY IS CAUSAL, FINITE, AND STATE DEPENDENT

We first measure the response to advocacy under matched initial conditions. For a fixed initial target fraction $x_0 = n_Z/N$, the susceptibility is

$$\chi(x_0) = \mathbb{E}[\Delta x \mid \text{ADV}, x_0] - \mathbb{E}[\Delta x \mid \text{NOOP}, x_0].$$

Holding the epistemic composition fixed, advocacy produces a positive shift toward the controller target at every tested state:

$$\chi(0.25) = 0.163, \quad \chi(0.375) = 0.100, \quad \chi(0.50) = 0.104.$$

The corresponding 95% matched-pair bootstrap intervals are $[0.119, 0.202]$, $[0.060, 0.144]$, and $[0.060, 0.148]$ (Fig. 1a). Thus the controller has a measurable causal influence even when the initial knowledge distribution is held fixed. The response is largest at the lowest tested target fraction and does not increase with additional pre-existing support over the tested range.

Microscopic controlled transitions provide a direct calibration of the finite-compliance description. Pooling the state-matched controlled updates gives an effective directional affinity $h_{\text{eff}} = 3.83$ and kinetic compliance $\gamma_{\text{eff}} = 0.372$. Substituting these into

$$\chi_{h,\gamma}(x) = [\sigma(h) - x] \left[1 - \left(1 - \frac{\gamma}{N} \right)^b \right]$$

gives the dashed finite-compliance reference in Fig. 1a. Because h_{eff} and γ_{eff} are estimated from the same microscopic trajectories, this is a calibrated reference rather than a parameter-free prediction. It nevertheless shows that separating directional bias from kinetic compliance removes much of the discrepancy with the idealized copying limit.

The exact theory also clarifies the role of compliance itself (Fig. 1b). Changing γ rescales the magnitude of the response while leaving the zero-response set point $x^* = \sigma(h)$ unchanged. This separation is useful empirically: h controls which direction is favored, whereas γ controls how often a controlled opportunity actually produces a revision.

6.2 STATE-LOCAL DYNAMICS REVEAL WHERE CONTROL IS EXPRESSED

The longer stochastic-feedback experiment gives access to the trajectory-level state dependence of control. Figure 2a shows the mean signed round-to-round drift $\mathbb{E}[\Delta x \mid x, b]$ for $q_c = 12$. Because the controller advocates an incorrect target, the uncontrolled epistemic dynamics usually drive x downward. Increasing b makes this correction progressively weaker, particularly at low and intermediate target support. This quantity is descriptive rather than causal: the matched experiment above remains the clean measurement of χ .

The controller action also leaves a strongly state-dependent information signature. We measure the reduced-state conditional mutual information

$$T_\pi(n) \equiv I(U_k; n_{k+1} \mid n_k = n),$$

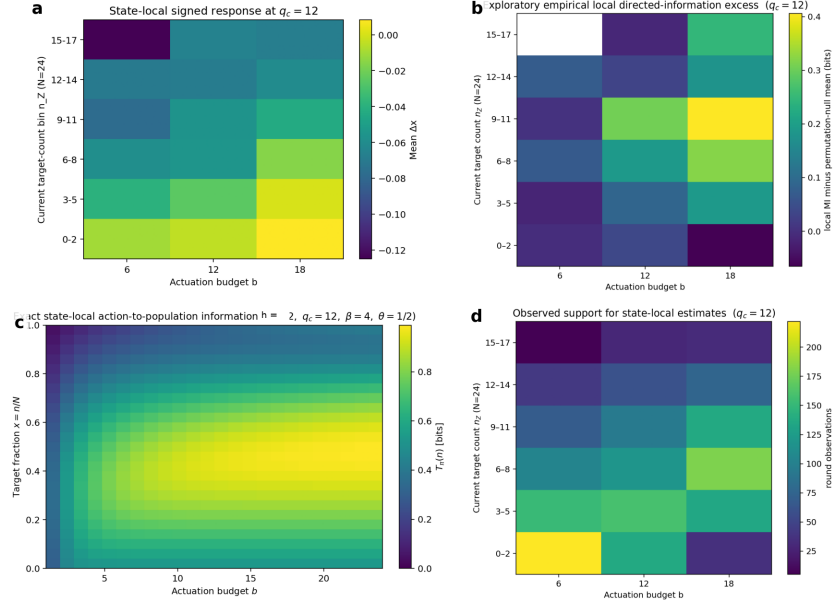


Figure 2: **State-local response and information.** (a) Empirical signed population drift as a function of target support and actuation budget at $q_c = 12$. (b) Empirical state-local action-to-population information above a within-state permutation null. (c) Exact theoretical state-local information landscape for the baseline single-affinity finite-compliance feedback controller. (d) Number of observed rounds supporting each empirical state bin.

the action-to-population information induced by the chosen feedback policy π . Here U_k is the policy-selected controller action and n_{k+1} is the population state after actuation. The empirical excess above a within-state permutation null is shown in Fig. 2b. At small b the signal is weak, whereas at $b = 18$ a clear interior band appears. The exact finite-state theory produces the same basic geometric mechanism (Fig. 2c): local information is high only where the action is not already deterministic and where the advocacy and NoOp transition kernels are sufficiently separated. The support map in Fig. 2d is included because the higher state bins are visited less often and should not be interpreted with the same confidence as densely sampled cells.

The robust conclusion is that control information is not uniform over population state space and stronger actuation creates an informative interior region. We do not interpret the precise ridge location as a phase boundary because n_Z is only a coarse-grained state.

6.3 FEEDBACK GAIN CONCENTRATES RATHER THAN SIMPLY INCREASES INFORMATION

The theory makes a more specific prediction about the feedback policy. The gain β controls how sharply the controller switches between advocacy and NoOp after sensing. At small β , the action remains uncertain over a wide range of states; at large β , the policy becomes nearly deterministic except close to its switching threshold. Consequently increasing β does not monotonically increase $T_{\pi}(n)$. Instead, the informative band narrows and moves toward the decision boundary (Fig. 3a).

The state-averaged theory curves make the same point in one dimension (Fig. 3b): for fixed actuation budget, larger β can reduce the average action-to-population information even though the controller is more decisive. This separates controller *decisiveness* from controller *information transmission* and suggests a focused validation rather than a broad search: $\beta \in \{1, 4, 8\}$ is sufficient to test whether the empirical information band sharpens as predicted.

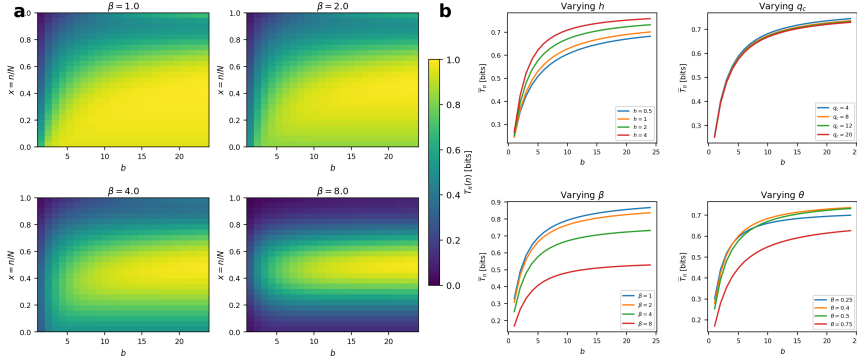


Figure 3: **Exact theory for feedback gain.** (a) State-local action-to-population information for increasing policy gain β . A sharper policy concentrates information around the switching region and suppresses it where the action saturates. (b) State-averaged local information versus actuation budget for the theory parameter sweeps; the β family shows that a more deterministic policy need not transmit more action information overall.

6.4 INFORMATION-RESPONSE EFFICIENCY INCREASES WITH ACTUATION

The information-response efficiency asks how much of the controller-dependent change in the full next-state distribution is expressed as a coherent shift of the mean target fraction. Using the bounded quantity defined in Section 5, we report the finite-horizon information-response efficiency $\eta_{\text{IR}}^{(H)}$, obtained as the ratio of the accumulated occupancy-weighted Pinsker response terms to the accumulated state-local action-to-population information over the evaluation horizon. The empirical values are small in absolute terms but systematic across the resource grid (Fig. 4a). They range from approximately 0.4% to 2.5% of the rigorous Pinsker bound and increase with b for every sensing budget. At $q_c = 12$, for example,

$$\eta_{\text{IR}} = 0.0043, 0.0135, 0.0228 \quad \text{for } b = 6, 12, 18.$$

Thus stronger actuation does more than increase the size of the population response: within this operating regime it also makes a larger fraction of the action-dependent distributional change visible as coherent mean motion. The small absolute values should not be interpreted as energetic losses; the Pinsker inequality supplies a conservative information-theoretic ceiling and need not be tight for these transition distributions.

6.5 THERMODYNAMIC EFFICIENCY EXPOSES THE SENSING-ACTUATION TRADE-OFF

The single-affinity path identity leads to a bounded thermodynamic control efficiency without introducing an opposing load. Removing the transient system entropy term from the finite-time path irreversibility gives

$$\Sigma - \Delta S_{\text{sys}} = hJ_c + I(n_k; Y_k),$$

and therefore

$$\eta_{\text{th}} = \frac{hJ_c}{\Sigma - \Delta S_{\text{sys}}} = \frac{hJ_c}{hJ_c + I(n_k; Y_k)} \leq 1.$$

The numerator is the affinity-weighted target-directed current; the denominator is the non-storage part of the finite-time thermodynamic accounting. Thus η_{th} asks what fraction of the sensing-plus-actuation expenditure is expressed as directed population motion. Appendix A.2 gives the complete path-measure derivation.

For empirical trajectories, we report the finite-horizon thermodynamic efficiency $\eta_{\text{th}}^{(H)}$. The affinity-weighted controlled current and scalar target-sensing information are accumulated cycle by cycle, and the efficiency ratio is formed only after this accumulation. We therefore do not average the cycle-wise efficiencies, and we do not compute sensing information from a single pooled occupancy distribution.

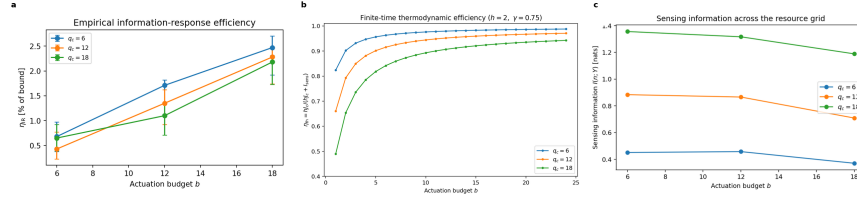


Figure 4: **The two control efficiencies and their resource dependence.** (a) Empirical information-response efficiency across the sensing-actuation grid. (b) Exact single-affinity thermodynamic efficiency $\eta_{th} = hJ_c / (hJ_c + I_{sens})$ as the actuation budget is varied for the three sensing budgets used experimentally. (c) Empirical sensing information, which rises strongly with q_c without a proportional increase in controlled current.

Figure 4b shows the exact single-affinity prediction for the same sensing budgets used experimentally, with $h = 2$, $\gamma = 0.75$, $\beta = 4$, $\theta = 1/2$, and $p_0(n) = \text{Binomial}(24, 0.25)$. At fixed q_c , the efficiency rises with actuation because the controlled current grows while the sensing information is fixed by the sensor channel. At fixed b , larger sensing budgets lower the efficiency unless the additional information produces a commensurate increase in current. For $b = 18$, for example,

$$\eta_{th} = 0.985, 0.964, 0.930 \quad \text{for } q_c = 6, 12, 18.$$

The result is therefore not that sensing is intrinsically undesirable, but that information that does not change the action-relevant population response is a thermodynamic overhead in this control objective.

The empirical sensing measurements support precisely this mechanism (Fig. 4c). Increasing q_c substantially raises $I(n_k; Y_k)$, whereas the calibrated controlled current changes much less across the same resource grid. Combined with the close agreement between the measured current and the finite-compliance current reported above, this yields a direct design prediction: in the present regime, adding actuation improves both η_{IR} and η_{th} , whereas additional sensing is efficient only when the policy can convert it into additional directed current.

Taken together, the results support a single resource-conversion picture of feedback control. Susceptibility identifies where the population can be moved; state-local information identifies where the action is dynamically visible; η_{IR} quantifies how coherently that information appears as mean motion; and η_{th} quantifies how efficiently the finite-time sensing-actuation budget is converted into affinity-weighted directed current. In the present regime, actuation is the dominant useful resource, while sensing displays clear diminishing returns. The theory additionally predicts that feedback gain β should reshape, rather than simply amplify, the information channel, providing a focused next validation of the framework.

7 CONCLUSION

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A THEORY AND DERIVATIONS

A.1 DERIVATION OF THE INFORMATION-RESPONSE BOUND

This appendix derives the bound

$$T_\pi(n) \geq \frac{2a_n(1-a_n)}{\ln 2} \chi_{h,\gamma}(n/N)^2 \quad (36)$$

used to define the information-response efficiency. The derivation proceeds in three steps: (i) expressing the conditional mutual information as a weighted Jensen-Shannon divergence, (ii) applying Pinsker’s inequality, and (iii) bounding the change in the mean target fraction by total variation.

Throughout the derivation, the current population state n is held fixed.

A.1.1 CONDITIONAL MUTUAL INFORMATION AS A JENSEN-SHANNON DIVERGENCE

Let the binary control variable be

$$U \in \{0, 1\},$$

where $U = 0$ denotes no intervention and $U = 1$ denotes advocacy of the controller target Z . At population state n , define

$$\Pr(U = 1 \mid n) = a_n, \quad \Pr(U = 0 \mid n) = 1 - a_n. \quad (37)$$

The corresponding next-round target-count distributions are

$$Q_0(n' \mid n) = \Pr(n' \mid U = 0, n), \quad (38)$$

$$Q_1(n' \mid n) = \Pr(n' \mid U = 1, n). \quad (39)$$

For notational convenience, while n is fixed, write

$$P(n') \equiv Q_0(n' \mid n), \quad Q(n') \equiv Q_1(n' \mid n), \quad a \equiv a_n. \quad (40)$$

The conditional mutual information between the control action and the next population state is, in the round-indexed notation of the main text,

$$T_\pi(n) \equiv I(U_k; n_{k+1} \mid n_k = n).$$

Holding $n_k = n$ fixed, we use the shorthand $U \equiv U_k$ and $n' \equiv n_{k+1}$ below. Thus

$$I(U; n' \mid n) = \sum_{u, n'} \Pr(u, n' \mid n) \log_2 \frac{\Pr(u, n' \mid n)}{\Pr(u \mid n) \Pr(n' \mid n)}. \quad (41)$$

Using the factorization

$$\Pr(u, n' | n) = \Pr(u | n) \Pr(n' | u, n),$$

the factor $\Pr(u | n)$ cancels inside the logarithm:

$$\begin{aligned} I(U; n' | n) &= \sum_{u, n'} \Pr(u | n) \Pr(n' | u, n) \log_2 \frac{\Pr(n' | u, n)}{\Pr(n' | n)} \\ &= \sum_u \Pr(u | n) D_{\text{KL}}(\Pr(n' | u, n) \| \Pr(n' | n)). \end{aligned} \quad (42)$$

We therefore need the next-state distribution when the value of U is not specified. By the law of total probability,

$$\begin{aligned} \Pr(n' | n) &= \sum_{u \in \{0,1\}} \Pr(u | n) \Pr(n' | u, n) \\ &= (1 - a)P(n') + aQ(n'). \end{aligned} \quad (43)$$

Define the mixture distribution

$$M(n') \equiv (1 - a)P(n') + aQ(n'). \quad (44)$$

Substituting the two possible actions into Eq. equation 42 gives

$$I(U; n' | n) = (1 - a)D_{\text{KL}}(P \| M) + aD_{\text{KL}}(Q \| M). \quad (45)$$

The weighted Jensen–Shannon divergence is defined by

$$\text{JS}_a(P, Q) \equiv (1 - a)D_{\text{KL}}(P \| M) + aD_{\text{KL}}(Q \| M), \quad M = (1 - a)P + aQ. \quad (46)$$

Hence, exactly,

$$I(U; n' | n) = \text{JS}_a(P, Q) \quad (47)$$

or, restoring the round-indexed notation,

$$T_\pi(n) \equiv I(U_k; n_{k+1} | n_k = n) = \text{JS}_{a_n}[Q_0(\cdot | n), Q_1(\cdot | n)]. \quad (48)$$

Up to this point, all relations are exact; no bound has yet been used.

A.1.2 PINSKER BOUND ON THE JENSEN–SHANNON DIVERGENCE

For two probability distributions A and B , define their total variation distance as

$$\text{TV}(A, B) \equiv \frac{1}{2} \sum_x |A(x) - B(x)|. \quad (49)$$

Pinsker’s inequality states that, when the Kullback–Leibler divergence is measured in bits,

$$D_{\text{KL}}(A \| B) \geq \frac{2}{\ln 2} \text{TV}(A, B)^2. \quad (50)$$

The factor $1/\ln 2$ appears because the standard Pinsker inequality is conventionally written for natural logarithms, whereas here the information is reported in bits.

Applying Eq. equation 50 separately to the two KL terms in Eq. equation 45 yields

$$T_\pi(n) \geq \frac{2}{\ln 2} [(1 - a) \text{TV}(P, M)^2 + a \text{TV}(Q, M)^2]. \quad (51)$$

The two total variation distances simplify because M is a mixture of P and Q . From

$$M - P = (1 - a)P + aQ - P = a(Q - P),$$

we obtain

$$\begin{aligned} \text{TV}(P, M) &= \frac{1}{2} \sum_{n'} |a(Q(n') - P(n'))| \\ &= a \text{TV}(P, Q). \end{aligned} \quad (52)$$

Similarly,

$$Q - M = Q - (1 - a)P - aQ = (1 - a)(Q - P),$$

and therefore

$$\text{TV}(Q, M) = (1 - a) \text{TV}(P, Q). \quad (53)$$

Substituting Eqs. equation 52–equation 53 into Eq. equation 51,

$$\begin{aligned} T_\pi(n) &\geq \frac{2}{\ln 2} [(1 - a)a^2 + a(1 - a)^2] \text{TV}(P, Q)^2 \\ &= \frac{2}{\ln 2} a(1 - a) \underbrace{[a + (1 - a)]}_{=1} \text{TV}(P, Q)^2. \end{aligned} \quad (54)$$

Thus

$$T_\pi(n) \geq \frac{2a(1 - a)}{\ln 2} \text{TV}(P, Q)^2. \quad (55)$$

Restoring $a = a_n$, $P = Q_0$, and $Q = Q_1$,

$$T_\pi(n) \geq \frac{2a_n(1 - a_n)}{\ln 2} \text{TV}(Q_0(\cdot | n), Q_1(\cdot | n))^2. \quad (56)$$

A.1.3 FROM TOTAL VARIATION TO THE MEAN TARGET-FRACTION RESPONSE

Let $n' \in \{0, \dots, N\}$ denote the next-round number of agents supporting the controller target. Define the corresponding target fraction

$$f(n') \equiv \frac{n'}{N}. \quad (57)$$

Since $0 \leq n' \leq N$,

$$0 \leq f(n') \leq 1. \quad (58)$$

Define the intervention-induced change in the mean target fraction by

$$\begin{aligned} \chi_{h,\gamma}(n/N) &\equiv \mathbb{E}_{Q_1} \left[\frac{n'}{N} \middle| n \right] - \mathbb{E}_{Q_0} \left[\frac{n'}{N} \middle| n \right] \\ &= \mathbb{E}_Q[f] - \mathbb{E}_P[f]. \end{aligned} \quad (59)$$

For completeness, we now show that any function satisfying $0 \leq f \leq 1$ obeys

$$|\mathbb{E}_Q[f] - \mathbb{E}_P[f]| \leq \text{TV}(P, Q). \quad (60)$$

Define

$$\Delta(n') \equiv Q(n') - P(n').$$

Since both P and Q are normalized,

$$\sum_{n'} \Delta(n') = 0. \quad (61)$$

Let

$$A_+ = \{n' : \Delta(n') \geq 0\}.$$

Then

$$\sum_{n' \in A_+} \Delta(n') = \text{TV}(P, Q). \quad (62)$$

Using Eq. equation 59,

$$\begin{aligned}\chi_{h,\gamma}(n/N) &= \sum_{n'} f(n')\Delta(n') \\ &= \sum_{n' \in A_+} f(n')\Delta(n') + \sum_{n' \notin A_+} f(n')\Delta(n').\end{aligned}\quad (63)$$

The second sum is non-positive because $f(n') \geq 0$ and $\Delta(n') < 0$ outside A_+ . Therefore

$$\begin{aligned}\chi_{h,\gamma}(n/N) &\leq \sum_{n' \in A_+} f(n')\Delta(n') \\ &\leq \sum_{n' \in A_+} \Delta(n') \\ &= \text{TV}(P, Q),\end{aligned}\quad (64)$$

where the second inequality uses $f(n') \leq 1$. Repeating the argument after exchanging P and Q gives the corresponding lower bound. Hence

$$|\chi_{h,\gamma}(n/N)| \leq \text{TV}(P, Q). \quad (65)$$

Squaring both sides,

$$\chi_{h,\gamma}(n/N)^2 \leq \text{TV}(P, Q)^2. \quad (66)$$

Finally, combining Eqs. equation 55 and equation 66,

$$\begin{aligned}T_\pi(n) &\geq \frac{2a_n(1-a_n)}{\ln 2} \text{TV}(Q_0(\cdot | n), Q_1(\cdot | n))^2 \\ &\geq \frac{2a_n(1-a_n)}{\ln 2} \chi_{h,\gamma}(n/N)^2.\end{aligned}\quad (67)$$

Therefore,

$$T_\pi(n) \geq \frac{2a_n(1-a_n)}{\ln 2} \chi_{h,\gamma}(n/N)^2. \quad (68)$$

For reference, the complete chain of relations is

$$\begin{aligned}T_\pi(n) &= I(U_k; n_{k+1} | n_k = n) \\ &= \text{JS}_{a_n}[Q_0(\cdot | n), Q_1(\cdot | n)] \\ &\geq \frac{2a_n(1-a_n)}{\ln 2} \text{TV}(Q_0(\cdot | n), Q_1(\cdot | n))^2 \\ &\geq \frac{2a_n(1-a_n)}{\ln 2} \chi_{h,\gamma}(n/N)^2.\end{aligned}\quad (69)$$

A.2 DERIVATION OF THE THERMODYNAMIC EFFICIENCY

This appendix derives the finite-time identity and the bounded thermodynamic control efficiency stated in Section 5.5. The calculation uses only the single directional affinity h , the kinetic compliance γ , and the finite sensor-policy channel.

A.2.1 CONTROLLED TRANSITION RATIO

For one controlled microscopic opportunity,

$$K_{h,\gamma}(n+1 | n) = \gamma \frac{N-n}{N} \sigma(h), \quad K_{h,\gamma}(n | n+1) = \gamma \frac{n+1}{N} \sigma(-h).$$

Since $\sigma(h)/\sigma(-h) = e^h$,

$$\frac{K_{h,\gamma}(n+1 | n)}{K_{h,\gamma}(n | n+1)} = \frac{N-n}{n+1} e^h. \quad (70)$$

The compliance factor cancels exactly: γ changes how frequently controlled revisions occur, but not their forward/reverse directional ratio. Define the combinatorial macrostate entropy

$$S_{\text{mix}}(n) = \ln \binom{N}{n}. \quad (71)$$

Because

$$S_{\text{mix}}(n+1) - S_{\text{mix}}(n) = \ln \frac{N-n}{n+1},$$

Eq. (70) becomes

$$\ln \frac{K_{h,\gamma}(n+1 | n)}{K_{h,\gamma}(n | n+1)} = S_{\text{mix}}(n+1) - S_{\text{mix}}(n) + h. \quad (72)$$

Equivalently, the positive weight

$$w_h(n) = \binom{N}{n} e^{hn} \quad (73)$$

satisfies

$$w_h(n) K_{h,\gamma}(m | n) = w_h(m) K_{h,\gamma}(n | m) \quad (74)$$

for every supported transition. Weighted reversibility is preserved under matrix multiplication. Hence for the b -opportunity advocacy kernel $Q_1 = K_{h,\gamma}^b$,

$$\boxed{\ln \frac{Q_1(m | n)}{Q_1(n | m)} = S_{\text{mix}}(m) - S_{\text{mix}}(n) + h(m - n).} \quad (75)$$

For NoOp, $Q_0(m | n) = \delta_{mn}$.

A.2.2 FORWARD AND REVERSE FEEDBACK PATHS

At feedback cycle k , let $p_k(n) = P(n_k = n)$. The forward path measure on (n, y, u, m) is

$$P_F^{(k)}(n, y, u, m) = p_k(n) S(y | n) \pi(u | y) Q_u(m | n). \quad (76)$$

Its final population marginal is

$$p_{k+1}(m) = \sum_{n,y,u} P_F^{(k)}(n, y, u, m), \quad (77)$$

and the forward sensor marginal is

$$p_{Y,k}(y) = \sum_n p_k(n) S(y | n). \quad (78)$$

We use the finite-time reverse reference

$$P_R^{(k)}(m, y, u, n) = p_{k+1}(m) p_{Y,k}(y) \pi(u | y) Q_u(n | m). \quad (79)$$

The reverse reference starts from the forward final ensemble p_{k+1} and therefore does not impose stationarity. The same policy factor $\pi(u | y)$ appears in both path weights and cancels from their ratio.

Define the stochastic path irreversibility

$$\Sigma_k^* \equiv \ln \frac{P_F^{(k)}(n, y, u, m)}{P_R^{(k)}(m, y, u, n)}. \quad (80)$$

Substitution gives

$$\Sigma_k^* = \ln \frac{p_k(n)}{p_{k+1}(m)} + \ln \frac{S(y | n)}{p_{Y,k}(y)} + \ln \frac{Q_u(m | n)}{Q_u(n | m)}. \quad (81)$$

Introduce the pointwise sensing information

$$i_k(n; y) = \ln \frac{S(y | n)}{p_{Y,k}(y)}, \quad (82)$$

the stochastic system entropy

$$s_{\text{sys}}(n; p) = -\ln p(n) + S_{\text{mix}}(n), \quad (83)$$

and the controlled target-count current increment

$$j_c = u(m - n). \quad (84)$$

For $u = 0$, $m = n$ and $j_c = 0$. For $u = 1$, Eq. (75) applies. Therefore

$$\Sigma_k^* = \Delta s_{\text{sys}} + h j_c + i_k(n; y). \quad (85)$$

A.2.3 FINITE-TIME SECOND LAW AND MEAN CONTROLLED CURRENT

Averaging Eq. (85) over the forward ensemble gives

$$\Sigma_k \equiv \mathbb{E}_F[\Sigma_k^*] = D_{\text{KL}}\left(P_F^{(k)} \| P_R^{(k)}\right) \geq 0. \quad (86)$$

The three average terms are

$$\Delta S_{\text{sys},k} = S_{\text{sys}}[p_{k+1}] - S_{\text{sys}}[p_k], \quad (87)$$

$$J_{c,k} = \mathbb{E}_F[j_c], \quad (88)$$

$$I_{\text{sens},k} = I(n_k; Y_k) = \sum_{n,y} p_k(n) S(y | n) \ln \frac{S(y | n)}{p_{Y,k}(y)}, \quad (89)$$

where

$$S_{\text{sys}}[p] = -\sum_n p(n) \ln p(n) + \sum_n p(n) \ln \binom{N}{n}. \quad (90)$$

Thus

$$\Delta S_{\text{sys},k} + h J_{c,k} + I_{\text{sens},k} = \Sigma_k \geq 0. \quad (91)$$

The current can be obtained directly from the susceptibility. Conditional on $n_k = n$ and advocacy,

$$\mathbb{E}[m - n | n, U = 1] = N \chi_{h,\gamma}(n/N) = N \Lambda_{b,\gamma} \left[\sigma(h) - \frac{n}{N} \right].$$

Advocacy occurs with probability a_n , so

$$J_{c,k} = N \Lambda_{b,\gamma} \sum_n p_k(n) a_n \left[\sigma(h) - \frac{n}{N} \right]. \quad (92)$$

This is the same current entering the affinity term of Eq. (91).

A.2.4 THERMODYNAMIC CONTROL EFFICIENCY

Equation (91) contains a finite-time boundary term, $\Delta S_{\text{sys},k}$, because the population ensemble need not be stationary. To isolate the part of path irreversibility associated with the feedback operation itself, define the non-storage control expenditure

$$C_{\text{th},k} \equiv \Sigma_k - \Delta S_{\text{sys},k}. \quad (93)$$

Using Eq. (91),

$$C_{\text{th},k} = h J_{c,k} + I_{\text{sens},k}. \quad (94)$$

The two contributions have distinct operational meanings. The first is the affinity-weighted directed population current produced by actuation; the second is the information acquired by sensing before the action is chosen.

For target-directed operation, $hJ_{c,k} \geq 0$. The fraction of the non-storage control expenditure expressed as directed current is therefore

$$\eta_{\text{th},k} \equiv \frac{hJ_{c,k}}{C_{\text{th},k}} = \frac{hJ_{c,k}}{\Sigma_k - \Delta S_{\text{sys},k}} = \frac{hJ_{c,k}}{hJ_{c,k} + I_{\text{sens},k}}. \quad (95)$$

Since $I_{\text{sens},k} \geq 0$,

$$0 \leq \eta_{\text{th},k} \leq 1 \quad (hJ_{c,k} \geq 0, C_{\text{th},k} > 0). \quad (96)$$

For a finite horizon, sum Eq. (91) over $k = 0, \dots, H-1$:

$$\sum_{k=0}^{H-1} \Delta S_{\text{sys},k} + h \sum_{k=0}^{H-1} J_{c,k} + \sum_{k=0}^{H-1} I_{\text{sens},k} = \sum_{k=0}^{H-1} \Sigma_k. \quad (97)$$

Because the system-entropy increments telescope,

$$S_{\text{sys}}[p_H] - S_{\text{sys}}[p_0] + hJ_c^{(H)} + I_{\text{sens}}^{(H)} = \Sigma^{(H)}, \quad (98)$$

where

$$J_c^{(H)} \equiv \sum_{k=0}^{H-1} J_{c,k}, \quad I_{\text{sens}}^{(H)} \equiv \sum_{k=0}^{H-1} I_{\text{sens},k}, \quad \Sigma^{(H)} \equiv \sum_{k=0}^{H-1} \Sigma_k. \quad (99)$$

The accumulated non-storage expenditure and corresponding efficiency are therefore

$$C_{\text{th}}^{(H)} = hJ_c^{(H)} + I_{\text{sens}}^{(H)}, \quad \eta_{\text{th}}^{(H)} = \frac{hJ_c^{(H)}}{C_{\text{th}}^{(H)}}. \quad (100)$$

In general,

$$\eta_{\text{th}}^{(H)} \neq \frac{1}{H} \sum_{k=0}^{H-1} \eta_{\text{th},k}, \quad (101)$$

because thermodynamic efficiency is a nonlinear ratio and the cycle-wise expenditures need not be equal.

At stationarity the boundary term vanishes and

$$\eta_{\text{th}} = \frac{hJ_c}{\Sigma}. \quad (102)$$

The finite-time expression is therefore not obtained by silently assuming $\Delta S_{\text{sys}} = 0$; rather, it removes that transient storage term explicitly before forming the ratio.

The efficiency makes the sensing–actuation trade-off transparent. Holding the sensor channel fixed, increasing actuation can raise hJ_c and drive η_{th} toward one. Conversely, increasing q_c can increase I_{sens} without a proportional increase in J_c , thereby reducing η_{th} . Kinetic compliance enters through J_c : a controller can have a large directional affinity but low thermodynamic efficiency when agents rarely comply and the resulting current remains small relative to the sensing information acquired.

B RELATIONAL-REASONING PROMPT

The population is driven by one versioned prompt family, `relational_public_ballot` version 1. Each call is compiled into two chat messages. The system message establishes the agent’s role, privacy boundary, task, call-specific option mapping, and currently known facts. The user message supplies the current vote and visible social source, then states the response contract. The model therefore receives only the state available to the focal agent at that microscopic update; it is never shown the global vote vector, other agents’ private facts, the ground-truth answer, or the controller’s sensor sample.

The display below reproduces one complete prompt from the audit of the minimal Study 04 inspection episode ($N = 24$, $q_c = 12$, $b = 12$). It is the first focal update of population round 1 for Agent 17. Typography is annotation only: **orange** marks values instantiated anew for the call, **green** marks the focal agent’s private state, and **blue** marks the visible social input. None of these colours or the “SYSTEM MESSAGE”/“USER MESSAGE” labels are sent to the model.

Audited focal-update prompt**SYSTEM MESSAGE**

You are **Agent 17**, one participant in a group reasoning problem.

Your goal is to identify the correct answer.

Participant identities persist across interactions.

Different participants know different facts about this problem. No single participant is guaranteed to know everything needed to answer, and some participants may have objectives that differ from yours. You do not know which participants, if any, have different objectives.

Make your own decision, using:

- the facts you currently know;
- your current position;
- the public positions of the participants shown below.

You see only the participants shown in this interaction, and of each one you see their vote and the fact they chose to share, if any.

Other participants will see the same of you: your vote and any fact you choose to share, and nothing else. Your reason is your own record: it is not shown to anyone.

QUESTION

Where is **Davi** relative to **Hira**?

The available answers are:

- **A) WEST**
- **B) EAST**
- **C) SOUTHWEST**

Exactly one of these answers is correct. Vote by its letter.

YOUR CURRENT KNOWLEDGE

You do not currently know any facts about this problem. You will have to rely on what other participants tell you.

USER MESSAGE**YOUR CURRENT POSITION**

Vote: **SOUTHWEST**

CURRENT SOCIAL INFORMATION

Agent 25

Vote: **B**

DECISION

Work out which option the facts available to you support, and vote for it.

Your reason should briefly explain your choice, for your own record.

Sharing a fact is the only way to pass information to other participants. You may share exactly one of the facts you currently know by giving its identifier in `shared_fact_id`, so that the participants who see your position can use it too. Use "none" if you prefer to share nothing.

You may share only a fact listed under **YOUR CURRENT KNOWLEDGE**, by its exact identifier. Do not invent facts, identifiers, or relationships that were not given to you.

Keep your reason to at most three sentences.

Return only valid JSON:

```
{
  "vote": "<A | B | C>",
  "reason": "<brief private reason>",
  "shared_fact_id": "<none>"
}
```

What changes between calls. The identity, query, available semantic answers, private facts, current vote, visible sources, and admissible fact identifiers are bound from the live game state. The semantic answers are independently permuted onto A/B/C for every call. Thus the **social vote B** above means **EAST** only in this instance; the persistent population state stores **EAST**, not the letter **B**. This prevents a stable display position from becoming an unintended convention.

Ordinary and controlled social inputs. An uncontrolled update shows the vote and, when supplied, one valid fact from an ordinary peer. In the audited example the visible source is the controller: the controller's semantic target **EAST** is rendered as **Vote: B** under this call's permutation. The prompt deliberately does not identify that source as a controller or authority. Under the

recommendation_only mode it also supplies no fact, so the intervention can change the public vote state but cannot directly enlarge the focal agent’s knowledge set.

Initialization variant. Before social interaction, each of the 24 agents receives the same role, task, privacy, fact-sharing, and JSON-contract blocks, but no current-position or social-information block. Instead, the prompt states that no other participant has yet declared a position and instructs the agent to decide from its local facts alone. The resulting 24 ballots define the initial population state.

Response and information boundary. The accepted response contains a display-letter vote, a private reason of at most three sentences, and either one fact identifier already in the agent’s knowledge set or none. The display letter is immediately mapped back to its semantic relation. The reason is retained for audit but never shown to another agent. Only the semantic vote and the explicitly selected fact may enter a later prompt, which makes the public decision channel and the epistemic transmission channel separately observable.

Audit check. The inspection episode recorded 72 distinct compiled prompt instances: 24 local-initialization prompts and 48 focal-update prompts (24 in each of two population rounds). All 72 responses passed validation without a retry or provider error. The prompt definition hash was constant across the episode, whereas every instance hash was unique, confirming that one fixed template was bound to 72 call-specific views.

B.1 EPISTEMIC CONDITIONS: RECEIVER DISPOSITION AND CONTROLLER EVIDENCE

The population game and the controller each expose one binary epistemic knob, and the two are set independently. The first, `receiver_epistemic_disposition`, decides what every agent is told about the social channel it listens to. The second, `controller_evidence_strategy`, decides which single fact the controller is allowed to cite when it speaks. They live at `game.options` and `control.options` respectively. Crossing them gives the four conditions used throughout Study 08, recorded in each round record as `derived_epistemic_condition`:

Condition	Receiver disposition	Controller evidence
<code>naive_neutral</code>	<code>naive</code>	<code>neutral</code>
<code>naive_strategic</code>	<code>naive</code>	<code>strategic</code>
<code>vigilant_neutral</code>	<code>vigilant</code>	<code>neutral</code>
<code>vigilant_strategic</code>	<code>vigilant</code>	<code>strategic</code>

The two axes act at different places in the loop. The receiver axis is a *text* intervention: it rewrites one fixed system block, identically for every agent and every call of an episode. The controller axis is a *content* intervention: it changes no template text at all, only which proposition of the frozen task the controller puts into its social slot. An agent therefore cannot read the controller’s strategy off the wording of its prompt, and the two manipulations remain separately attributable.

Receiver epistemic disposition. This axis substitutes the `social_environment` block, the second system block of the manifest. Both variants are bound as *fixed*: they are part of the episode’s prompt definition rather than of any per-call view, so within one cell every one of the episode’s prompts carries the same wording. `naive` states only that identities persist, and says nothing about whether a visible participant is worth believing. `vigilant` adds an explicit instruction to discount a bare recommendation and to check the evidence actually in hand. The two blocks are reproduced below exactly as sent.

social_environment — naive

Participant identities persist across interactions.

social_environment — vigilant

Participant identities persist across interactions.

Some participants may have objectives that differ from yours. Do not accept a recommendation merely because another participant gives it. Evaluate whether the explicit evidence you actually have supports the conclusion, and remain aware that a source may present only part of the relevant information.

Nothing else in the prompt moves with this axis. In the audited grid the block is estimated at 6 tokens under *naive* and 55 under *vigilant*, carrying the whole per-prompt difference of 160 versus 209 estimated tokens. The block manifest — names, order, roles, and bindings — is byte-identical across all four cells, because it records prompt *structure*; the condition is pinned by the *receiver_epistemic_disposition* field of each cell’s resolved configuration and is visible in the rendered text itself.

The audited Study 04 prompt displayed earlier in this appendix predates this factorization. It shows the older combined block, which asserts both distributed information and possible objective divergence in one paragraph. The factorized axis separates those two claims, and Study 08 uses only the *naive/vigilant* pair above.

Controller evidence strategy. Evidence reaches the population only in the controller’s message mode *recommendation_plus_fact*, in which the controller occupies its social slot with both a vote and exactly one fact of the frozen task. That fact is chosen once, before the episode runs, by a deterministic rule that never consults a model:

- *neutral* takes the first fact in the task’s frozen fact order. The choice ignores which relation the controller advocates, so the cited evidence is fixed by the task alone.
- *strategic* takes the admissible fact whose relation points most nearly in the advocated direction. Each of the eight compass relations has a symbolic displacement vector; the controller scores every fact by the cosine between its relation vector and the advocated relation’s vector, discards every fact scoring zero or below, and cites the survivor with the highest score, breaking ties toward the earlier fact. If no fact has positive alignment the configuration is rejected at launch rather than silently advocating without evidence.

strategic is thus the selective-evidence arm: the controller still never fabricates anything, since only real facts of the task are admissible, but it is free to cite the true statement that best flatters its target and to stay silent about the rest. *neutral* removes that freedom while holding the recommendation, the budget, and the number of exposures fixed, so the difference between the arms isolates the value of evidence *selection*.

Worked example. Study 08’s task `task_0001` asks where Zora is relative to Feni, with frozen fact order (f_1, f_2, f_3): f_1 “Zora is south of Maro” (SOUTH), f_2 “Maro is west of Feni” (WEST), and the distractor f_3 “Garó is north of Belo” (NORTH), which mentions neither query entity. The correct relation is SOUTHWEST. The rules then resolve as follows.

Advocated relation	<i>neutral</i>	<i>strategic</i>
SOUTHWEST (correct)	f_1 (SOUTH)	f_1 (SOUTH)
NORTHWEST	f_1 (SOUTH)	f_2 (WEST)
NORTH	f_1 (SOUTH)	f_3 (NORTH)

The last row is the informative one. Advocating NORTH, the *neutral* controller must cite f_1 , whose southward relation openly contradicts its own recommendation. The *strategic* controller instead cites f_3 : a true statement, pointing north, about two entities irrelevant to the question. Under *naive* the agent is given no reason to notice the substitution; under *vigilant* it is told in so many words to check whether the evidence it holds supports the conclusion.

Audited grid. The four conditions were run once end to end against a live model. Each cell is one episode of $N = 6$ agents over 4 population rounds with a single visible social slot ($K = 1$), sensor sample $q_c = 3$, exact budget $b = 3$, target *correct*, *recommendation_plus_fact*,

on gwdg/openai-gpt-oss-120b at temperature 0 with seed 20260828. Each cell issued 30 calls — 6 local-initialization ballots and $6 \times 4 = 24$ focal updates — for 120 in total, matching the recorded request count exactly, so no response required a retry.

In this particular task the two evidence arms coincide: the advocated relation is the correct one, SOUTHWEST, and the tie between $\mathfrak{f}1$ and $\mathfrak{f}2$ — both at cosine $1/\sqrt{2}$ — resolves to the earlier fact, which is also the neutral rule’s choice. All four cells therefore exposed $\mathfrak{f}1$, and the evidence axis is degenerate here by construction. The grid is a prompt- and plumbing-level audit of the four conditions, not an estimate of their effect; the axis separates only on tasks and targets where alignment discriminates, as the two lower rows of the worked example show.

C CONTROLLED SUSCEPTIBILITY STUDY

C.1 STATE-MATCHED RESPONSE TO ADVOCACY

We isolate the one-round behavioral response of the population to controller advocacy while holding the epistemic population fixed. The experiment uses $N = 24$ agents, $q = 1$ social interaction, reasoning depth $L = 2$, support redundancy $r = 6$, recommendation-only control, and an incorrect controller target Z . We fix the sensing and actuation budgets to $q_c = b = 18$ and explicitly initialize three target-support levels,

$$x_0 \in \{0.25, 0.375, 0.50\}, \quad n_Z(0) \in \{6, 9, 12\}.$$

Within each task, agent identities and initial knowledge sets $K_i(0)$ are identical across all three conditions, giving $\kappa_0 = 0.25$ and $\phi_0 = 0$. Target supporters are selected deterministically and balanced across the populated knowledge strata, with nested sets $Z_6 \subset Z_9 \subset Z_{12}$; all remaining agents are initialized at the correct relation. Thus only the initial social support for Z changes across conditions.

For each x_0 , we compare two deterministic one-round interventions: ADVOCATE, in which the controller advocates for Z at exactly $b = 18$ controlled update positions, and NOOP, in which it does not intervene. The two arms use matched episode seeds. We evaluate two tasks with ten matched repetitions per task, yielding 120 one-round episodes. The sensor remains active for logging, but it does not determine the forced action.

The primary quantity is the signed intervention response

$$\chi(x_0) = \mathbb{E}[x_1 - x_0 \mid \text{ADV}, x_0] - \mathbb{E}[x_1 - x_0 \mid \text{NOOP}, x_0], \quad (103)$$

with 95% confidence intervals from 1000 bootstrap resamples of matched episode pairs. Since the action is deterministic within each arm, this experiment estimates a response contrast rather than an action-to-population mutual information.

The measured responses are $\chi(0.25) = 0.163$ (95% CI $[0.119, 0.202]$), $\chi(0.375) = 0.100$ ($[0.060, 0.144]$), and $\chi(0.50) = 0.104$ ($[0.060, 0.148]$). The controller therefore produces a positive causal shift toward its target at all three matched states. However, the largest response occurs at the lowest tested target fraction, and increasing the initial target support from $x_0 = 0.375$ to 0.50 produces no additional gain. The exact $q = 1$ reference predicts the same qualitative ordering—largest leverage at low target support—but substantially larger responses (0.401, 0.334, and 0.268, respectively). Thus, in this controlled one-round setting, LLM advocacy is effective but markedly weaker than the idealized copying reference, and there is no evidence that additional pre-existing social support amplifies the intervention response.

C.2 EMPIRICAL CALIBRATION OF DIRECTIONAL AFFINITY AND COMPLIANCE

The single-affinity theory describes a controlled interaction using a directional bias h and a kinetic compliance γ . To connect these quantities to the LLM dynamics, we use the microscopic transitions observed in the state-matched intervention experiment. We coarse-grain each focal agent’s vote into two states: supporting the controller target Z , or not supporting Z . Among controlled updates, define

$$p_+ = P(\text{non-}Z \rightarrow Z \mid \text{controlled}), \quad p_- = P(Z \rightarrow \text{non-}Z \mid \text{controlled}).$$

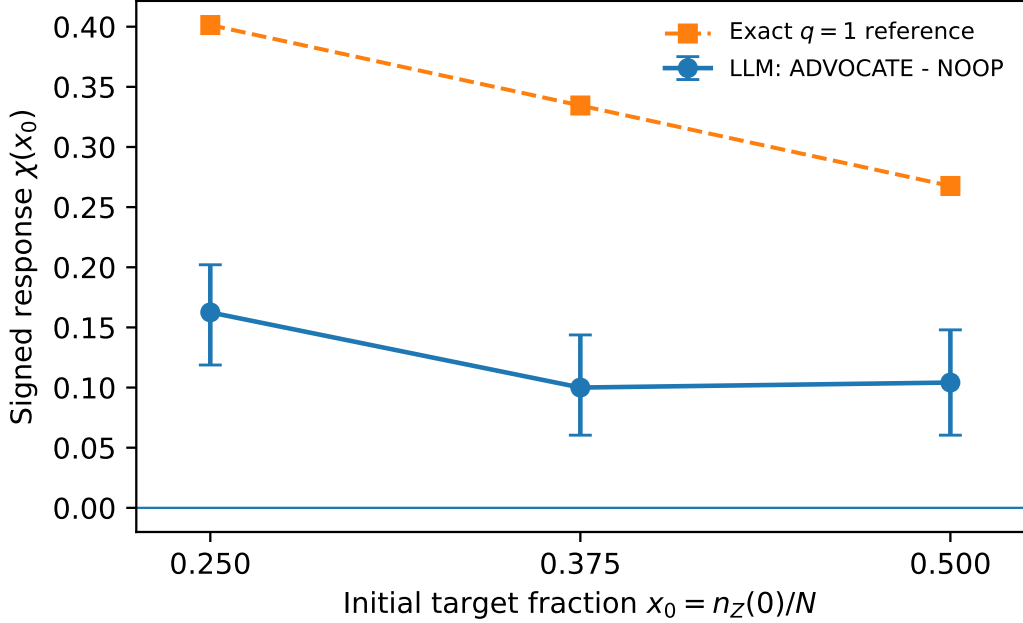


Figure 5: One-round state-matched response to advocacy. Points show the empirical ADVOCATE-minus-NOOP response with 95% matched-pair bootstrap intervals. The dashed curve is the exact $q = 1$ reference at $N = 24$ and $b = 18$, $\chi_{q=1}(x_0) = (1 - x_0)[1 - (1 - 1/N)^b]$. Advocacy produces a positive response at every imposed initial target fraction, while the response does not increase with pre-existing target support over the tested range.

Table 1: Microscopic transitions observed at controlled update positions.

Transition	Eligible updates	Observed transitions	Probability
non- $Z \rightarrow Z$	572	208	$p_+ = 0.364$
$Z \rightarrow$ non- Z	508	4	$p_- = 0.0079$

Under the finite-compliance kernel,

$$p_+ = \gamma\sigma(h), \quad p_- = \gamma\sigma(-h).$$

Therefore the two parameters are directly identifiable:

$$\gamma_{\text{eff}} = p_+ + p_-, \quad h_{\text{eff}} = \ln \frac{p_+}{p_-}. \quad (104)$$

The sum measures how readily a controlled opportunity produces any revision, whereas the ratio measures the direction favored conditional on revision.

Pooling all controlled updates gives

$$h_{\text{eff}} = 3.83, \quad \gamma_{\text{eff}} = 0.372,$$

with forward-to-reverse transition ratio $e^{h_{\text{eff}}} \simeq 46.2$. Thus motion toward the controller target is strongly favored among revisions, while most controlled opportunities still leave the focal vote unchanged. A stratified episode-level bootstrap gives the approximate 95% interval

$$h_{\text{eff}} \in [3.06, 4.88].$$

No further decomposition of h_{eff} is required by the finite-time thermodynamic theory. The same measured affinity enters the susceptibility, the controlled current, and the path identity $\Delta S_{\text{sys}} + hJ_c + I(n_k; Y_k) = \Sigma$.

Table 2: Empirical quantities connecting microscopic LLM dynamics to the single-affinity finite-compliance description.

Quantity	Estimate	Interpretation
h_{eff}	3.83	directional affinity
95% CI for h_{eff}	[3.06, 4.88]	bootstrap uncertainty
$e^{h_{\text{eff}}}$	46.2	forward/reverse transition ratio
γ_{eff}	0.372	probability scale for actual revision

D RESOURCE-CONSTRAINED CONTROL DESIGN

The theory in Section 5 separates three questions that are often conflated in controller design: how strongly the population can be moved, how coherently controller influence is converted into that motion, and how efficiently the finite-time sensing–actuation budget is converted into directed current. This appendix turns susceptibility, the two bounded efficiencies, and an explicit operational interaction cost into a practical design prescription.

D.1 MINIMUM-COST CONTROL AT PRESCRIBED RESPONSE

Let

$$\varphi = (q_c, b, \theta, \beta, \dots)$$

denote the tunable feedback-controller configuration, and let $\rho(n)$ be the population-state occupancy over the evaluation window. The microscopic channel parameters h and γ are treated separately: unless the implementation exposes a mechanism that changes them directly, they are calibrated properties of the population response rather than free design knobs.

Since the state-local susceptibility is defined conditionally on advocacy, the expected controller-induced response per round is

$$R(\varphi; h, \gamma) = \sum_{n=0}^N \rho(n) a_n(\varphi) \chi_{h,\gamma}\left(\frac{n}{N}\right). \quad (105)$$

A direct operational interaction cost is

$$C_{\text{op}}(\varphi) = q_c + b \sum_{n=0}^N \rho(n) a_n(\varphi), \quad (106)$$

where the first term counts the agents sensed every round and the second counts the expected number of controlled microscopic opportunities.

The most direct controller-design problem is therefore not to maximize an efficiency in isolation, but to achieve a prescribed behavioral response at minimum operational cost,

$$\varphi^* = \arg \min_{\varphi} C_{\text{op}}(\varphi) \quad \text{subject to} \quad R(\varphi; h, \gamma) \geq R_{\text{req}}. \quad (107)$$

Equivalently, for a fixed interaction budget one may maximize the achievable response,

$$\varphi^* = \arg \max_{\varphi} R(\varphi; h, \gamma) \quad \text{subject to} \quad C_{\text{op}}(\varphi) \leq C_{\text{max}}. \quad (108)$$

These formulations make the practical question explicit: once the required response has been fixed, how much sensing and actuation can be removed before performance falls below the desired level?

For the exact finite-compliance $q = 1$ reference,

$$\chi_{h,\gamma}(x) = [\sigma(h) - x] \Lambda_{b,\gamma}, \quad \Lambda_{b,\gamma} = 1 - \left(1 - \frac{\gamma}{N}\right)^b. \quad (109)$$

The marginal gain from one additional controlled opportunity is

$$\Lambda_{b+1,\gamma} - \Lambda_{b,\gamma} = \frac{\gamma}{N} \left(1 - \frac{\gamma}{N}\right)^b, \quad (110)$$

which decreases monotonically with b . Thus stronger actuation continues to increase the mean response toward the set point, but each additional controlled opportunity contributes less than the previous one. The same logic applies to sensing: increasing q_c is useful only insofar as the improved state estimate changes the policy in a way that improves the required response.

D.2 HOW THE TWO EFFICIENCIES GUIDE FEASIBLE DESIGNS

The operational problem above specifies how to purchase less control. The two bounded efficiencies from Section 5 then determine whether a feasible controller uses its influence and thermodynamic budget well.

The information–response efficiency is

$$\eta_{\text{IR}}(n) = \frac{2a_n(1-a_n)\chi_{h,\gamma}(n/N)^2}{(\ln 2)T_\pi(n)} \leq 1, \quad (111)$$

with $T_\pi(n) = I(U_k; n_{k+1} | n_k = n)$. It measures whether the action-dependent change in the full next-state distribution is expressed as a coherent shift in mean target fraction. Among settings with comparable cost and response, one should prefer regimes in which η_{IR} remains high over the states actually visited.

The thermodynamic efficiency is

$$\eta_{\text{th},k} = \frac{hJ_{c,k}}{\Sigma_k - \Delta S_{\text{sys},k}} = \frac{hJ_{c,k}}{hJ_{c,k} + I(n_k; Y_k)} \leq 1. \quad (112)$$

Here $\Delta S_{\text{sys},k}$ is the transient storage term and $\Sigma_k - \Delta S_{\text{sys},k}$ is the non-storage finite-time control expenditure. The efficiency therefore reports the fraction of the generalized sensing–actuation expenditure expressed as affinity-weighted directed current. Its derivation from the forward/reverse path ratio is given in Appendix A.2.

Two practical rules follow directly. First, more sensing is not automatically better: if increasing q_c raises $I(n_k; Y_k)$ without a proportional increase in hJ_c , it lowers η_{th} . Second, more actuation is useful only while it continues to raise the required response; the exact susceptibility already shows diminishing marginal gains through $\Lambda_{b+1,\gamma} - \Lambda_{b,\gamma}$.

Taken together, the design prescription is

- (i) specify the minimum acceptable response or current,
 - (ii) minimize operational cost subject to that requirement,
 - (iii) use η_{IR} to reject incoherent action channels,
 - (iv) among feasible controllers, prefer larger η_{th} ,
 - (v) retain h and γ as separate directional and kinetic diagnostics.
- (113)

Thus susceptibility determines where control has leverage, the operational cost determines how much control must be purchased, η_{IR} determines whether controller influence becomes coherent mean motion, and η_{th} determines whether sensing and actuation are converted efficiently into directed current.

D.3 FINITE-TIME AND NONSTATIONARY INTERPRETATION

No stationarity assumption is required in the construction above. Let

$$\rho_k(n) = \Pr(n_k = n) \quad (114)$$

denote the transient population distribution at round k . Over a finite horizon of H rounds, the occupancy entering Eqs. (105)–(106) can be interpreted as the finite-time visitation distribution

$$\bar{\rho}_H(n) = \frac{1}{H} \sum_{k=0}^{H-1} \rho_k(n). \quad (115)$$

The visitation distribution $\bar{\rho}_H$ can be used directly for observables that are linear in the occupancy, such as the expected controlled response and current. For example,

$$\frac{1}{H} \sum_{k=0}^{H-1} J_{c,k} = N \sum_n \bar{\rho}_H(n) a_n \chi(n). \quad (116)$$

Sensing information is different because mutual information is nonlinear in the state distribution. In general,

$$I_{\text{sens}}[\bar{\rho}_H] \neq \frac{1}{H} \sum_{k=0}^{H-1} I_{\text{sens}}[\rho_k]. \quad (117)$$

Finite-horizon thermodynamic accounting therefore evaluates $I_{\text{sens},k}$ from each transient distribution ρ_k and accumulates the resulting cycle-wise contributions before forming $\eta_{\text{th}}^{(H)}$. This preserves the finite-time interpretation without imposing stationarity or replacing the transient sequence by a single pooled ensemble.

Hence the operational design problem averages over the transient states actually visited during the experiment rather than over an equilibrium distribution. Likewise, Eq. (91) retains the finite-time boundary contribution $\Delta S_{\text{sys},k}$, which accounts for the fact that the population ensemble changes during the feedback cycle. A stationary distribution is therefore only a special long-time limit of the present formulation, not an assumption required by it.